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Java Example of Meta-Schema v1

Validate payloads with Amazon Product Type Definitions meta-schema instances.

Example Validator Implementation for Java

For Java applications, the [networknt/json-schema-validator](#) library supports JSON Schema Draft 2019-09 and custom vocabularies. The following example demonstrates how to utilize the [networknt/json-schema-validator](#) library to validate payloads with instances of the *Amazon Product Type Definition Meta-Schema*. There is no requirement to use this specific library or the example implementation. Amazon does not provide technical support for third-party JSON Schema libraries and this is provided as an example only.

Schema Configuration

When using the [networknt/json-schema-validator](#) to validate instances of the *Amazon Product Type Definition Meta Schema* with custom vocabulary, the meta-schema is configured as part of the `JsonSchemaFactory`.

Constants:

Java

```
// $id of the Amazon Product Type Definition Meta-Schema.
String schemaId = "https://schemas.amazon.com/selling-partners/definitions/product-types/meta-schema/v1";

// Local copy of the Amazon Product Type Definition Meta-Schema.
String metaSchemaPath = "./amazon-product-type-definition-meta-schema-v1.json";

// Local copy of an instance of the Amazon Product Type Definition Meta-Schema.
String luggageSchemaPath = "./luggage.json";

// Keywords that are informational only and do not require validation.
List<String> nonValidatingKeywords = ImmutableList.of("editable", "enumNames");
```

Configure Meta-Schema:

Java

```
// Standard JSON Schema 2019-09 that Amazon Product Type Definition Meta-Schema extends from.
JsonMetaSchema standardMetaSchema = JsonMetaSchema.getV201909();

// Build Amazon Product Type Definition Meta Schema with the standard JSON Schema 2019-09 as the blueprint.
// Register custom keyword validation classes (see below).
JsonMetaSchema metaSchema = JsonMetaSchema.builder(SCHEMA_ID, standardMetaSchema)
```

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Vendor Central URLs

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Building Robust Amazon SP-API Applications

SP-API SDKs

Java Example of Meta-Schema v1

```
.addKeywords(NON_VALIDATING_KEYWORDS.stream().map(NonValidationKeyword::new)
    .collect(Collectors.toSet()))
.addKeyword(new MaxUniqueItemsKeyword())
.addKeyword(new MaxUtf8ByteLengthKeyword())
.addKeyword(new MinUtf8ByteLengthKeyword())
.build();
```

Build JsonSchemaFactory:

```
Java
// URIFetcher to route meta-schema references to local copy.
URIFetcher uriFetcher = uri -> {
    // Use the local copy of the meta-schema instead of retrieving from the
    web.
    if (schemaId.equalsIgnoreCase(uri.toString())) {
        return Files.newInputStream(Paths.get(metaSchemaPath));
    }

    // Default to the existing fetcher for other schemas.
    return new URLFetcher().fetch(uri);
};

// Build the JsonSchemaFactory.
JsonSchemaFactory schemaFactory = new JsonSchemaFactory.Builder()
    .defaultMetaSchemaURI(schemaId)
    .addMetaSchema(standardMetaSchema)
    .addMetaSchema(metaSchema)
    .uriFetcher(uriFetcher, "https")
    .build();

// Create the JsonSchema instance.
JsonSchema luggageSchema = schemaFactory.getSchema(new
String(Files.readAllBytes(Paths.get(luggageSchemaPath))));
```

Payload Validation

With an instance of the *Amazon Product Type Definition Meta-Schema* loaded as a `JsonSchema` instance, payloads can be validated using the instance.

```
Java
// Create a JsonNode for the payload (this can be constructed in code or
read from a file).
JsonNode payload = new ObjectMapper().readValue(new File("./payload.json"),
JsonNode.class);

// Validate the payload and get any resulting validation messages.
Set<ValidationMessage> messages = luggageSchema.validate(payload);
```

If no validation messages are returned, validation passed. Otherwise, inspect the validation messages to identify errors with the payload.

Keyword Validation

The [networknt/json-schema-validator](https://github.com/networknt/json-schema-validator) supports validating custom vocabulary by using classes that extend the `AbstractKeyword` class and provide the validation logic.

Refer to <https://github.com/networknt/json-schema-validator/blob/master/doc/validators.md>.

The following examples illustrate extensions of the `AbstractKeyword` class that validate the custom vocabulary in instances of the *Amazon Product Type Definition Meta-Schema*.

MaxUniqueItemsKeyword Class

Java

		<pre>import com.fasterxml.jackson.databind.JsonNode; import com.fasterxml.jackson.databind.node.JsonNodeType; import com.google.common.collect.Maps; import com.google.common.collect.Sets; import com.google.common.collect.Streams; import com.networknt.schema.AbstractJsonValidator; import com.networknt.schema.AbstractKeyword; import com.networknt.schema.CustomErrorMessageType; import com.networknt.schema.JsonSchema; import com.networknt.schema.JsonValidator; import com.networknt.schema.ValidationContext; import com.networknt.schema.ValidationMessage; import org.apache.commons.lang3.StringUtils; import java.text.MessageFormat; import java.util.Iterator; import java.util.Map; import java.util.Optional; import java.util.Set; import java.util.stream.Collectors; /** * Example validator for the "maxUniqueItems" keyword. */ public class MaxUniqueItemsKeyword extends AbstractKeyword { private static final MessageFormat ERROR_MESSAGE_FORMAT = new MessageFormat("Each combination of selector " + "values may only occur {1} times. The following selector value combination occurs too many times: {2}"); private static final String KEYWORD = "maxUniqueItems"; private static final String SELECTORS = "selectors"; public MaxUniqueItemsKeyword() { super(KEYWORD); } @Override public JsonValidator newValidator(String schemaPath, JsonNode schemaNode, JsonSchema parentSchema, ValidationContext validationContext) { // Only process if the provided schema value is a number. if (!JsonNodeType.NUMBER.equals(schemaNode.getNodeType())) { return null; } int maxUniqueItems = schemaNode.asInt(); // Get the selector properties configured on the scheme element, if // they exist. Otherwise, this validator // defaults to using all properties. Set<String> selectors = getSelectorProperties(parentSchema); return new AbstractJsonValidator(this.getValue()) { @Override public Set<ValidationMessage> validate(JsonNode node, JsonNode rootNode, String at) { // Only process if the node is an array, as selectors and // unique items do not apply to other data // types. // Create a property-value map of each items properties // (selectors) and count the number of // occurrences for each combination. } } } }</pre>
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```
package com.amazon.spucs.tests.keywords;

import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.node.JsonNodeType;
import com.networknt.schema.AbstractJsonValidator;
import com.networknt.schema.AbstractKeyword;
import com.networknt.schema.CustomErrorMessageType;
import com.networknt.schema.JsonSchema;
import com.networknt.schema.JsonValidator;
import com.networknt.schema.ValidationContext;
import com.networknt.schema.ValidationMessage;

import java.nio.charset.StandardCharsets;
import java.text.MessageFormat;
import java.util.Set;

/**
 * Example validator for the "maxUtf8ByteLength" keyword.
 */
public class MaxUtf8ByteLengthKeyword extends AbstractKeyword {

    private static final MessageFormat ERROR_MESSAGE_FORMAT =
        new MessageFormat("Value must be less than or equal {1} bytes in length.");

    private static final String KEYWORD = "maxUtf8ByteLength";

    public MaxUtf8ByteLengthKeyword() {
        super(KEYWORD);
    }

    @Override
    public JsonValidator newValidator(String schemaPath, JsonNode schemaNode, JsonSchema parentSchema, ValidationContext validationContext) {

        // Only process if the provided schema value is a number.
        if (!JsonNodeType.NUMBER.equals(schemaNode.getNodeType())) {
            return null;
        }

        int maxUtf8ByteLength = schemaNode.asInt();

        return new AbstractJsonValidator(this.getValue()) {

            @Override
            public Set<ValidationMessage> validate(JsonNode node, JsonNode rootNode, String at) {

                // Get the value as a string and evaluate its length in bytes.
                String value = node.asText();
                if (value.getBytes(StandardCharsets.UTF_8).length > maxUtf8ByteLength) {
                    return fail(CustomErrorMessageType.of(KEYWORD, ERROR_MESSAGE_FORMAT), at, Integer.toString(maxUtf8ByteLength));
                }
                return pass();
            }
        };
    }
}
```

MinUtf8ByteLengthKeyword Class

Java

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```
package com.amazon.spucs.tests.keywords;

import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.node.JsonNodeType;
import com.networknt.schema.AbstractJsonValidator;
import com.networknt.schema.AbstractKeyword;
import com.networknt.schema.CustomErrorMessageType;
import com.networknt.schema.JsonSchema;
import com.networknt.schema.JsonValidator;
import com.networknt.schema.ValidationContext;
import com.networknt.schema.ValidationMessage;

import java.nio.charset.StandardCharsets;
import java.text.MessageFormat;
import java.util.Set;

/**
 * Example validator for the "minUtf8ByteLength" keyword.
 */
public class MinUtf8ByteLengthKeyword extends AbstractKeyword {

    private static final MessageFormat ERROR_MESSAGE_FORMAT =
        new MessageFormat("Value must be greater than or equal {1} bytes in length.");

    private static final String KEYWORD = "minUtf8ByteLength";

    public MinUtf8ByteLengthKeyword() {
        super(KEYWORD);
    }

    @Override
    public JsonValidator newValidator(String schemaPath, JsonNode schemaNode, JsonSchema parentSchema, ValidationContext validationContext) {
        // Only process if the provided schema value is a number.
        if (!JsonNodeType.NUMBER.equals(schemaNode.getNodeType())) {
            return null;
        }

        int minUtf8ByteLength = schemaNode.asInt();

        return new AbstractJsonValidator(this.getValue()) {

            @Override
            public Set<ValidationMessage> validate(JsonNode node, JsonNode rootNode, String at) {

                // Get the value as a string and evaluate its length in bytes.

                String value = node.asText();
                if (value.getBytes(StandardCharsets.UTF_8).length < minUtf8ByteLength) {
                    return fail(CustomErrorMessageType.of(KEYWORD, ERROR_MESSAGE_FORMAT), at, Integer.toString(minUtf8ByteLength));
                }
                return pass();
            }
        };
    }
}
```

APPLICATION MANAGEMENT API

Application Management API >

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